are kept in hot water and the loose ends of the thread are caught by hand. Threads from several cocoons are wound together on spinning wheels (Charakhas) to form the reels of raw silk. Only about one-half of the silk of each cocoon is reelable, the remainder is used as a silk waste and formed into spun silk. Raw silk thus obtained is processed through several treatments to bring about the luster on the thread.



New silkworm diet produces coloured silk. The Institute of Materials Research and Engineering (IMRE) in Singapore has developed a way to replace the traditional dying process necessary to make coloured silk. A simple dietary change (feeding a diet of mulberries treated with fluorescent dye) for the silkworm larva and they are able to produce silk in a variety of colors. The colour directly integrated into the fibers.



Uses of Silk

1. Silk fibers are utilized in preparing silk clothes. Silk fibers are now combined with other natural or synthetic fibers to manufacture clothes like **Teri-Silk, Cot-Silk** etc. Silk is dyed and printed to prepare ornamented fabrics. They are generally made from Eri-silk or spun silk.
2. Silk is used in industries and for military purposes.
3. It is used in the manufacture of fishing fibers, parachutes, cartridge bags, insulation coils for telephone, wireless receivers, tyres of racing cars, filter fibres, in medical dressings and as suture materials.

Diseases and Pests of Silkworm:

The profitable silk industry is threatened by various diseases caused by the virus, fungal, bacterial and protozoan infections but also by insect predators, birds and other higher animals. Ants, crows, kites, rats, feed upon silk worms thereby causing a great loss to silk industry. Pebrine, is a dangerous disease to in silkworms and the causative organism is *Nosema bombycis* , a protozoan. This silkworm disease is transmitted through the egg of the mother silkworm and also through ingestion of contaminated food. Flacherie generally occurs in the mature larvae and is caused mainly by bacteria like *Streptococcus* and *Staphylococcus*. Grasserie is a most dominant and serious viral disease. It is caused by *Bombyx mori* nuclear polyhedrosis virus (BmNPV) a *Baculovirus*, which belongs to sub group 'A' of the Baculoviridae. Among the fungal diseases, white muscardine is common. This disease is caused by fungus *Beauveria bassiana*.

TN_GOVT_IX_Zoology_Ch13.indd 244 09-01-2020 5.21.31 PM